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Access to Hydroxy-Functionalized Polypropylene through Coordination Polymerization

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Abstract: *Preparation of polyethylenes containing hydroxy groups has been already industrialized through radical copolymerization under harsh conditions followed by alcoholysis. By contrast, hydroxy-functionalized polypropylene has proven a rather challenging goal in polymer science. Propylene can't be polymerized through a radical mechanism, and its coordination copolymerization with polar monomers is frustrated by catalyst poisoning. Herein, we report a new strategy to reach this target. The coordination polymerization of allenes by rare-earth-metal precursors affords pure 1,2-regulated polyallenes, which are facilely transformed into poly(allyl alcohol) analogues by subsequent hydroboration/oxidation. Strikingly, the copolymerization of allenes and propylene gives unprecedented hydroxy-functionalized polypropylene after post-polymerization modification. Mechanistic elucidation by DFT simulation suggests kinetic rather than thermodynamic control.*

The synthesis of hydroxy-functionalized (co)polymers, in particular polyolefins, has been a major focus of polymer scientists because the hydroxy group endows the materials with hydrophilicity and excellent surface properties, and furthermore, it is a reactive group that enables access to other functionalities. For instance, polyethylene (PE) is the most famous and widely applied hydrophobic plastic,[1] whilst its hydroxy-functionalized derivatives such as poly(vinyl alcohol) (PVA) and poly(ethylene-vinyl alcohol) (EVOH) possess rather unique properties and applications in different fields.[2] To date the copolymerization of ethylene with a broad scope of functional vinyl monomers has been thoroughly investigated and achieved great successes,[3] except for hydroxy functionalization. Fortunately, radical polymerization of vinyl acetate or copolymerization with ethylene under high temperature and pressure followed by post-polymerization alcoholysis affords PVA and EVOH, which is a rather efficient industrialized method.[4] By contrast, synthesis of the corresponding hydroxy-functionalized polypropylene (PP) from the direct polymerization of allylic alcohol (AA) or its copolymerization with propylene poses serious problems. Although allyl alcohol (AA) is a stable chemical, it is hard to propagate into a high molecular weight PAA through radical polymerization even with the assistance of a Lewis acid;[5] furthermore anionic and coordination catalysts react with the hydroxy group rather than the C=C bond. A more efficient manner of obtaining a linear PAA with high molecular weight (ca. 30 kDa) and a block copolymer with styrene P(AA-St) was reported recently by Hillmyer and co-workers through reversible addition–fragmentation chain transfer (RAFT) radical polymerization of the Boc-protected (bis(tert-butyloxy carbonate)) vinyl ester or vinyl amide followed by post-polymerization reduction.[6] The copolymerization of propene with allyl acetate is another way to reach PP that includes hydroxy functionalities after hydrolysis, but the insertion rate of allyl acetate was poor.[7] Therefore, the coordination polymerization approach used to fabricate PP hasn't been applied to prepare PAA or hydroxy-containing PP, since even the Boc-protected polar monomers are poisonous to the employed coordination catalysts.

Allene compounds are widely employed as building blocks in the field of organic synthesis due to their highly active cumulative double bonds.[8] They have also attracted the attention of polymer scientists because the corresponding polymers possess double bonds that can be transformed into other functionalities, for example, through hydrogenation,[9] epoxidation,[10] and hydrosilylation,[11] in post-polymerization modification steps. However, the high activity of the cumulative double bonds causes difficulty in controlling the polymer microstructures through either radical or cationic polymerization.[12] Well-controlled polymerization of allenes is achieved by using coordination catalysts, of which the Ti-based systems[10], η^3-allyl $NiOCOCF_3$,[13] and the rare-earth-metal catalysts[14] display almost perfect 2,3-selectivity. Nevertheless, the resultant polymers contain internal C=C double bonds that are chemically stable and resistant to further transformation. On the other hand, 1,2-selective polymerization of allenes gives polymers with the active terminal C=C double bonds, which is sterically disfavored.

Herein, we report the highly active and distinguished 1,2-selective coordination polymerization of allene monomers

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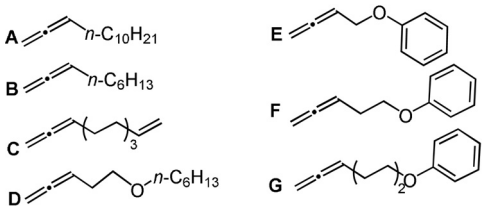
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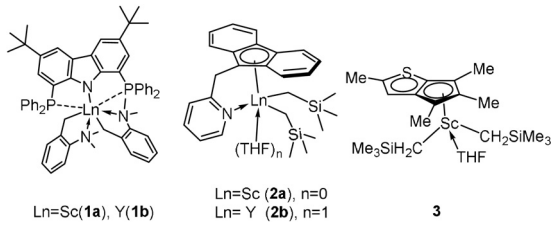
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(Scheme 1) and its copolymerization with propylene by using the rare-earth metal precursors (Scheme 2). The resultant poly(allene)s have high molecular weights and are facilely transformed into analogous PAAs through hydroboration/oxidation reactions under mild conditions. Strikingly, the copolymerization of propylene and allene monomers is also successful to afford unprecedented hydroxy-functionalized PP after post-polymerization modification. The mechanism for the activity and 1,2-selectivity was elucidated through DFT simulation, which indicated that the process is under kinetic rather than thermodynamic control.



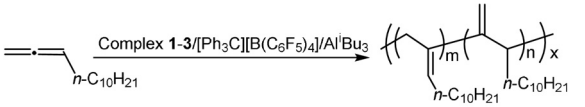
Scheme 1. Allene monomers.



Scheme 2. Rare-earth-metal precursors **1**–**3**.

PNP-carbazolide scandium (**1a**) and yttrium (**1b**) bis(alkyl) complexes have been reported to show excellent activity and perfect *cis*-1,4-selectivity in the activation of $[Ph_3C][B(C_6F_5)_4]$ for the polymerization of conjugated diene monomers.[15] Both were tested for catalyzing *n*-decylallene polymerization (Table 1, entries 1 and 2) but were completely inert. Then, constrained geometry configuration(CGC) catalysts, namely fluorenyl rare-earth metal bis(alkyl) complexes with methylene pyridine side arms [(Flu-CH_2-Py)Ln-$(CH_2SiMe_3)_2(THF)_n$ (Ln = Sc, $n=0$, **2a**; Y, $n=1$, **2b**)] in combination with $[Ph_3C][B(C_6F_5)_4]$ and Al^iBu_3 were employed. The polymerization of *n*-decylallene with yttrium precursor **2b** afforded only trace amounts of polymer, in contrast to its behavior in phenylallene polymerization, where it was reported previously to exhibit high activity and pure 2,3-selectivity.[14] Upon switching to the more Lewis acidic scandium precursor **2a**, the polymerization proceeded efficiently and reached 90% conversion in 24 h. The isolated polymer showed 1,2-regularity (72%; Table 1, entries 3 and 4). Therefore, we turned to the less bulky half-sandwich thiophene-fused-cyclopentadienyl scandium complex **3**, which is highly active for ethylene polymerization.[16] The polymerization went smoothly to reach 92.2% conversion under the same conditions to with precursors **2** and gave a poly(*n*-decylallene) with a moderate molecular weight ($M_n = 1.8\times10^4$; Table 1, entry 5). The ^{1}H NMR spectrum of the resultant polymer (Figure 1, top) shows two groups of broad peaks at $\delta = 5.21$ and 2.8 ppm corresponding to the terminal vinyl protons H_a and methine proton H_b on the 1,2-regulated polymer chains, while the broad resonances within the range of $\delta = 2.0$ to 0.5 ppm are attributed to the alkyl $C_{10}H_{21}$ protons. The peak at $\delta = 5.63$ ppm corresponding to the internal methylene protons H_c is rather weak. The ratio of the integral intensities of H_a:H_c indicates a 94% 1,2-tacticity, which is higher than that obtained from any other catalysts reported to date.

Table 1: Polymerization of *n*-decylallene catalyzed by complexes **1**–**3**.[a]



Entry	Cat	Conv.(%)	1,2%[b]	M_n[c]$\times10^{-4}$	PDI[c]	T_g[d] [°C]	T_m[d] [°C]
1	**1a**	trace	–	–	–	–	–
2	**1b**	0	–	–	–	–	–
3[e]	**2a**	90	72	1.3	1.39	−27	–
4[e]	**2b**	trace	–	–	–	–	–
5[f]	**3**	92.2	94	1.8	1.47	−27	–

[a] Conditions: Cat 10 μmol, Mon/Cat = 100, $[Ph_3C][B(C_6F_5)_4]$ 10 μmol, Toluene 0.2 mL, $T = 20$°C, Time 24 h. [b] Measured by ^{1}H NMR in $C_6D_4Cl_2$ at 100°C. [c] Determined by GPC in $C_6H_3Cl_3$ at 150°C against polystyrene standard. [d] Determined by DSC. [e] Al^iBu_3 10 μmol. [f] Al^iBu_3 20 μmol.

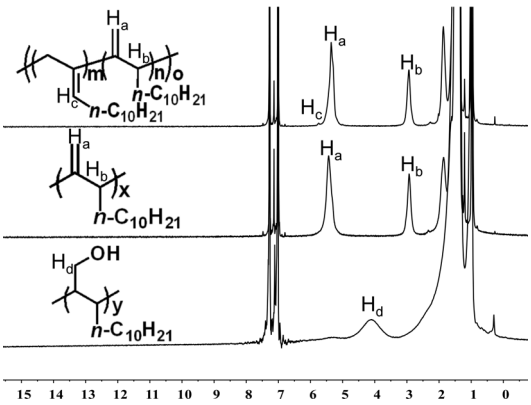
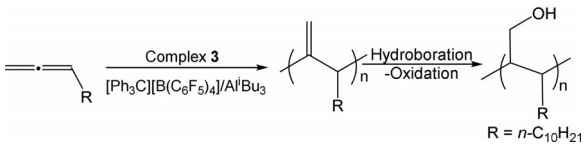


Figure 1. ^{1}H NMR spectrum of poly(*n*-decylallene) obtained at 20°C (top, Table 1, entry 5), −40°C (middle, Table 2, entry 6) and hydroxylated polymer (bottom) (400 MHz, $C_6D_4Cl_2$, 100°C).

Intrigued by these results, we investigated in detail the catalytic performance of complex **3** under various conditions. Representative data are shown in Table 2. Upon changing the *n*-decylallene/Sc ratio from 100:1 to 500:1, the molecular weight of the resultant poly(*n*-decylallene) increased correspondingly from $M_n = 1.8\times10^4$ to 3.6×10^4 while the molecular-weight distribution was maintained (Table 1, entry 5; Table 2, entries 1 and 2). This indicates that the polymerization is controllable and the active species is single-sited, which avoids any side reactions and crosslinking products. Notably, the addition of Al^iBu_3 (Al:Sc = 10:1) to the above binary system [Sc]/$[Ph_3C][B(C_6F_5)_4]$ did not show any obvious

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Table 2: Polymerization of allene derivatives catalyzed by the thiophene-fused cyclopentadiene-ligated scandium complex **3** followed by post-functionalization.[a]



Entry	Mon.	Mon./Cat.	T [°C]	Time [h]	Conv. [%]	1,2 %[b]	M_n[c] × 10^{-4}	PDI[c]	T_g[d] [°C]	T_m[d] [°C]
1	**A**	200	20	24	94.6	94	2.2	1.67	−27	–
2	**A**	500	20	24	82.4	94	3.6	1.67	−27	–
3	**A**[e]	100	20	24	94.8	94	1.9	1.60	−27	–
4	**A**	100	0	24	72.8	96	3.7	1.30	−27	–
5	**A**	100	−20	36	86.6	98	4.6	1.25	−27	39
6	**A**	100	−40	36	85.1	>99	7.6	2.07	–	39
7	**A**	500	−40	96	91.2	>99	25.6	2.62	–	39
8	**B**	100	20	24	83.8	94	1.1	1.71	−10	–
9	**B**	100	−40	36	81.9	>99	5.1	1.56	–	–
10	**C**	100	20	24	89.1	94	1.4	2.19	−32	–
11	**C**	100	−40	36	82.4	>99	5.0	1.82	–	–
12	**D**	100	20	24	0	–	–	–	–	–
13	**E**	100	20	24	0	–	–	–	–	–
14	**F**[f]	50	20	96	41.9	13.0	0.5	1.65	23	–
15	**G**	100	20	36	78.7	84.0	0.8	1.85	−2	–
16	**G**	100	−40	24	65.8	>99	2.8	1.60	−2	–

[a] Conditions used for polymerization: [Sc] 10 μmol, $[Ph_3C][B(C_6F_5)_4]$ 10 μmol, Al^iBu_3 20 μmol, Toluene 0.2 mL. The data in table are for the polymers before post-functionalization. [b] Measured by ^1H NMR in $C_6D_4Cl_2$ at 100 °C [c] Determined by GPC in $C_6H_3Cl_3$ at 150 °C against polystyrene standard. [d] Determined by DSC. [e] [Sc] 10 μmol, $[Ph_3C][B(C_6F_5)_4]$ 10 μmol, Al^iBu_3 100 μmol. [f] [Sc] 20 μmol.

influence on the catalytic performances (Table 2, entry 3). Upon lowering the polymerization temperature to 0 °C, a higher-molecular-weight polymer with a distinguished 1,2-regularity ($M_n = 3.7 \times 10^4$, 96 %) was afforded. Upon further decreasing the temperature to −20 °C and −40 °C, the polymerization still proceeded to give an over 85 % conversion, albeit at a prolonged polymerization time. Meanwhile, the 1,2-selectivity increased strikingly to over 99 % (Figure 1, middle). In the ^{13}C NMR spectrum, the terminal vinyl carbon C_a shows a resonance at $\delta = 109$ ppm and the ternary carbon C_b displays a sharp singlet at $\delta = 152$ ppm, whilst the methine carbon C_c appears at $\delta = 51$ ppm. There are no resonances from the other regularities, thus indicating almost perfect 1,2-regularity (Figure 2), which is further confirmed by the HSQC spectrum analysis (Figure S39). Compared to the 2,3-selectivity observed for allene polymerization, 1,2-selectivity means a spatially disfavored route, therefore, the high recorded value of 1,2-selectivity is, as far as we are aware, indeed unconventional. The physical properties of poly(*n*-decylallene) products are strongly dependent on the microstructure. The polymer with a 94 % 1,2-tacticity possesses a T_g value of −27 °C without a melting point (T_m), while that bearing an over 99 % 1,2-tacticity gives a melting point $T_m = 39$ °C owing to its more regulated microstructure, albeit containing long soft alkyl side chains (Figure 3).

The distinguished performances of complex **3** towards *n*-decylallene drove us to explore the polymerizations of allene derivatives bearing different substituents. Monomer **B**, which contains an *n*-hexyl group (Scheme 1), was polymerized to give poly(*n*-hexylallene) with the identical 1,2-regularity as poly(n-decylallene), thus suggesting that the length of the side alkyl chain has no obvious influence on the selectivity

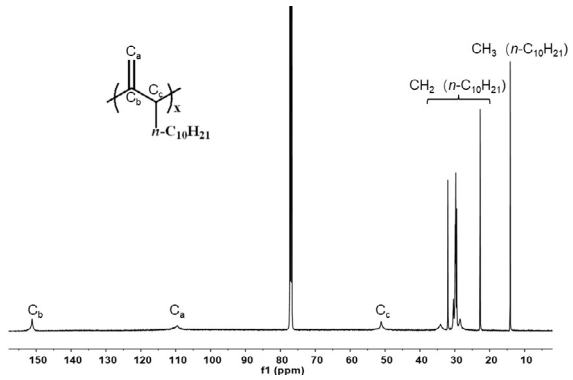


Figure 2. ^{13}C NMR spectrum of poly(*n*-decylallene) obtained at −40 °C (Table 2, entry 6; 125 MHz, $CDCl_3$, 25 °C).

(Table 2, entries 8 and 9). The polymerization of monomer **C**, which possesses a terminal C=C double bond, exhibited regioselectivity at the allene unit exclusively, thus indicating the high reactivity of the cumulative double bonds compared to discrete double bonds (Table 2, entries 10 and 11). To investigate the influence of polar substituents, monomers **D**, **E**, **F**, and **G** were synthesized (Scheme 1). Monomer **D** has a propyloxo fragment attached to the cumulative double bonds. During the polymerization, the oxygen atom is geometrically available to back-bite the active Sc^{3+} ion together with the cumulative unit, thereby generating a strong chelate that retards the polymerization (Scheme S1 in the Supporting Information). Replacing the *n*-C_6H_{13} species in **D** with a phenyl group is anticipated to prevent oxygen from poisoning the active species, and monomer **F** indeed could be polymerized sluggishly to reach 42 % conversion in 72 h

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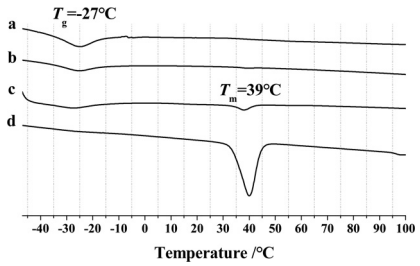


Figure 3. DSC traces of poly(*n*-decylallene)s with different 1,2-tacticities. a) 1,2% = 94% (Table 1, entry 5); b) 1,2% = 96% (Table 2, entry 4); c) 1,2% = 98% (Table 2, entry 5). d) 1,2% > 99% (Table 2, entry 6).

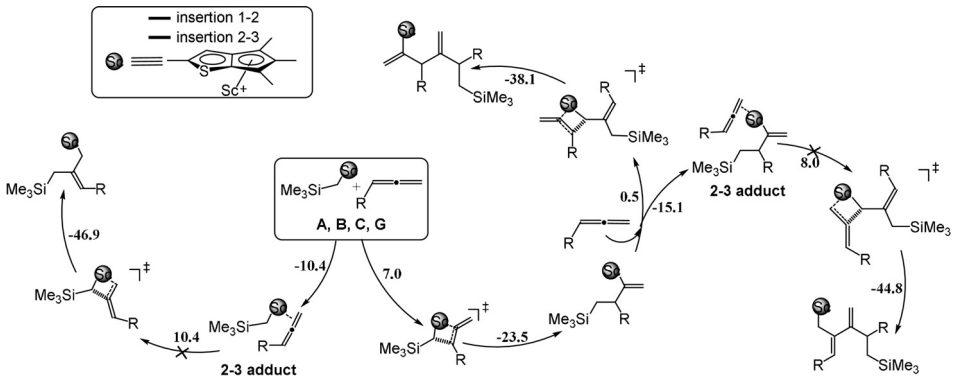
albeit with a rather low 1,2-selectivity (13.0%). This might be attributed to weak back-biting chelation of the phenoxy group to the active metal center to give a crowded and variable coordination–insertion environment. Monomer **E**, which bears the methylene unit between the cumulative double bonds and phenoxy group, was completely inert since it coordinates to the active species and then isomerizes to 1-phenoxy-1,3-butadiene through 1,3-hydrogen shift.[17] Strikingly, monomer **G**, which contains a butyl spacer, polymerized fluently to generate pure 1,2-tacticity (Table 2, entries 12–16).

The mechanism of this highly active and chemoselective (1,2 vs. 2,3) performance for the polymerization of allenes **A**, **B**, **C**, or **G** was then elucidated by DFT calculations at the (B3PW91) level based on catalysts **2a** and **3**. In the following, only insertion of monomer **B** is considered as representative of the allene polymerization. Experimentally, complex **2a** appears to be less 1,2-selective than catalyst **3**. The two first insertions (1,2 and 2,3) were computed for catalyst **2a** (see Figure S64). It was found that there is clear kinetic competition between the two insertions, which is in line with the moderate selectivity of this complex for the 1,2-insertion. Indeed, the first insertion seems to be 1,2 (barrier of 10.0 vs. 13.6 kcal mol^{-1}), whereas the second one might be a 2,3 (11.9 kcal mol^{-1} vs. 6.9 kcal mol^{-1}). On the other hand, the reactivity of complex **3** seems very different (Scheme 3). Interestingly, the 1,2-insertion appears to be kinetically always more favorable than the 2,3 one (barrier of 7.0 vs. 10.4 kcal mol^{-1} for the first insertion and of 0.5 vs. 8.0 kcal mol^{-1} for the second one). This is in line with the greater selectivity for the 1,2-insertion observed experimentally. This selectivity appears to be due to the fact that no stable 1,2-adduct can be formed whereas a stable 2,3 adduct can. The stability of the latter therefore reduces the 2,3 reactivity. However, it should be noted that 1,2-selectivity appears to be kinetically driven since the 2,3-insertion product, which appears to be a π-allyl, is thermodynamically favored. Therefore, at high temperature, formation of the thermodynamic product should be observed, thereby reducing the 1,2-selectivity, whereas the 1,2-selectivity should be greater at low temperature (low energy demand for the reaction). This is in perfect agreement with the experimental results (Table 2, entries 8 and 9). Therefore, the reaction sequence is controlled by the kinetically facile 1,2-insertion but not by the coordination of the monomer to the metal center. This finding is quite unusual for classical coordination-insertion mechanisms and is associated with steric repulsion between the growing polymer chain and the allene, which prevents the formation of a 1,2-adduct.

Hydrolysis of the obtained polymers was performed through hydroboration with BH_3 followed by oxidation with H_2O_2 under mild conditions. In the ^{1}H NMR spectrum, the peak at $\delta = 5.21$ ppm assigned to the terminal vinyl protons is almost invisible, while the broad resonance at $\delta = 3.75$ ppm is attributed to the methylene CH_2 group adjacent to the hydroxy group, thus suggesting the rather high efficiency of this post-polymerization modification (Figure 1, bottom). In the FTIR (cm^{-1}) trace, the characteristic C=C stretching vibration at 1648 arising from 1,2-regulated poly(alkylallene) is nearly absent while a strong and broad resonance at 3300 cm^{-1} is attributed to the stretching vibration of the newly generated O–H bond (Figure S61).

The surface properties of the nonpolar poly(alkylallene) is greatly improved after hydrolysis to decyl substituted PAA, which gives a smaller static water contact angle (WCA) of 96.2° as compared to 107.7° for poly(alkylallene).[18]

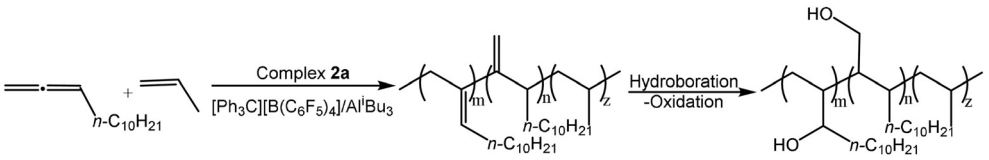
Compared to the extensive investigations on functionalizing polyethylene through coordination polymerization of ethylene with polar monomers,[3] that for polypropylene is rather rare. One reason is that polypropylene cannot be prepared through a radical mechanism, which avoids radical copolymerization with vinyl acetylacetate; another is efficient catalysts for propylene coordination polymerization are very limited compared to those for ethylene polymerization, thus those that can survive in the copolymerization with polar monomers are scarce. The success of highly 1,2-selective polymerization of the nonpolar allene compounds prompted us to attempt their copolymerization with propylene using the excellent precursor **3**. Unfortunately, only trace amounts of polymer were obtained, which might be due to intense competition between the two monomers. Surprisingly, upon activa-



Scheme 3. Proposed mechanism for the polymerization of alkylallene monomers **A**, **B**, **C**, and **G** by catalyst **3** [computed enthalpy of monomer **B** is provided (kcal mol^{-1})].

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Table 3: Copolymerization of *n*-decylallene and propylene by complex **2a** followed by post-functional.[a]



Entry	Propylene [atm]	*T* [°C]	Time [h]	Yield [g]	Allene %[b]	1,2 %[b]	M_n[c] $\times 10^{-4}$	PDI	T_g [°C]	T_m[d] [°C]
1	1	20	0.5	0.10	26.5	79	1.0	1.4	−40	–
2	1	20	1	0.26	10.3	79	0.8	1.6	−21	–
3	2	20	1	0.71	7.6	79	0.8	1.5	−17	–
4	4	20	1	1.18	5.1	79	0.9	1.4	−15	–
5	1	0	3	0.36	9.5	84	1.2	1.4	−19	3
6[e]	1	20	1	0.19	0	–	0.8	1.5	−6	–

[a] Conditions used for polymerization: [Sc] 10 μmol, Allene/[Sc]/[Ph_3C][$B(C_6F_5)_4$] = 100:1:1 (mol mol^{-1}), toluene, 2 mL, 1 h, r.t. The data in table are for the copolymers before post-functionalization. [b] Measured by ^{1}H NMR in $CDCl_3$ at 25 °C. [c] Determined by GPC in THF at 40 °C against polystyrene standard. [d] Determined by DSC. [e] Homopolymerization of propylene.

tion with [Ph_3C][$B(C_6F_5)_4$], complex **2a** successfully mediated the copolymerization of allene monomer **A** and propylene with high activity comparable to that of propylene homopolymerization and rather high 1,2-selectivity (80 %). Meanwhile the allene incorporation was sensitive to the propylene fraction in the reaction vessel, which increased with a decrease in propylene pressure and reaction temperature to reach up to 26.5 % (Table 3). This meant that after post-functionalization, a polypropylene containing as high as a 26.5 % molar fraction of hydroxy groups was realized for the first time.

In summary, we have demonstrated that the unconventional highly 1,2-selective polymerization of polar and nonpolar alkylallene compounds can be achieved with high activity by using rare-earth-metal catalytic systems. These alkyl-substituted allene monomers are sterically demanding, thus, from a catalytic precursor viewpoint, a more open coordination sphere is a precondition to show activity for allene polymerization. Half-sandwich thiophene-fused cyclopentadienyl-ligated rare-earth metal precursors are sterically less bulky than the pyridinyl methylene fluorenyl and the tridentate PNP ligated rare-earth-metal precursors, and exhibit higher activity. From the allene monomer viewpoint, a long alkyl chain or a terminal double bond does not prohibit polymerization but a polar oxygen atom can terminate the polymerization by chelating to the active metal center. With respect to chemoselectivity, the 1,2-selective polymerization is under kinetic rather than thermodynamic control. This work opens a new avenue to hydroxy modification of nonpolar polymers through copolymerization synthons while avoiding the problem of catalysts being poisoned by the polar groups. The copolymerizations of allenes with styrene and conjugated dienes and other simple olefins is currently under investigation.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: allenes · copolymerization · hydroxylated polypropylenes · polar monomers · rare-earth metals

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